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MAC-S^{3}Robo: Mamba-Driven Contrastive Semi-Supervised Surgical Robot Segmentation for Minimally Invasive Surgery

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ABSTRACT Accurate and reliable surgical robot segmentation during endoscopic surgery is critical for improving robotic perception, task automation, and patient safety. Current deep learning approaches rely heavily on supervised training, requiring extensive, manually annotated datasets, which are costly and difficult to obtain in surgical contexts. To address this challenge, we propose MAC-S^{3}Robo, a Mamba-driven contrastive semi-supervised segmentation framework tailored to robotic and endoscopic surgical segmentation. Specifically, it integrates two Mamba-based branches, cross pseudo-label supervision on unlabelled frames, and labelled supervision within a unified optimisation objective. Within this framework, a Mamba-based U-shaped encoder–decoder replaces U-Net convolutional blocks with visual state-space (VSS) blocks while preserving the multi-scale design and identity skip connections, enabling stronger modelling of long-range dependencies and global context. Contrastive self-supervision is integrated into the same objective to strengthen pixel-level representations and stabilise features, benefiting the semi-supervised setting. Extensive experiments on the EndoVis 2017 and 2018 benchmarks demonstrate that MAC-S^{3}Robo achieves the best mean Dice among semi-supervised baselines at 10% labelled data on both datasets and maintains consistent gains as the labelled ratio increases. The implementation is available at https://github.com/ziyangwang007/CV-SSL-Robot.

INDEX TERMS Surgical robotics, Endoscopic vision, Instrument and scene segmentation, Mamba, Self-supervised learning, Semi-supervised learning.

I. Introduction

Minimally invasive surgery (MIS) has transformed clinical practice by reducing trauma, shortening recovery time, and improving postoperative outcomes. Robotic assistance further extends MIS by providing tremor suppression, enhanced dexterity, and precise instrument control within constrained workspaces. To fully realise these advantages, reliable computer vision is paramount: dense and temporally consistent perception of surgical scenes, thereby facilitating higher levels of autonomy, contextual assistance, and robust human–robot collaboration [1], [2]. Accordingly, we focus on dense, per-pixel segmentation of both robotic instruments and surrounding scene elements as the core perception primitive for surgical robotics. However, the primary impediment is the scarcity and expense of annotations with pixel-level accuracy: curating dense masks is labour intensive and difficult to harmonise across centres, even as hospitals accumulate large volumes of raw endoscopic video that remain underexploited for training. Meanwhile, clinical centres continuously acquire extensive endoscopic footage, yet converting this raw material into reliable supervision at large scale remains prohibitive given the time and expertise required for labelling at the frame level with pixel-level accuracy [3]. This persistent imbalance between scarce labels and abundant raw data has therefore become a central bottleneck for deploying robust segmentation systems in practice. This motivates learning regimes that reduce reliance on dense labels by exploiting the abundant, routinely captured unlabelled endoscopic video.

Robotic instrument segmentation is central to surgical perception, supporting downstream tasks such as tool tracking, depth and scene reconstruction, augmented reality overlays, and skill assessment [4]. Early convolutional neural network (CNN)-based approaches leveraged encoder–decoder designs (e.g., U-Net variants) [5], [6] with hand-crafted augmentations, while more recent methods such as TraSeTR [7] and ISINet [8] introduced transformer-based, attention-based, or temporal consistency cues to model long-range dependencies. On the supervised side, architectures such as Attention U-Net [9] and TransUNet [10] strengthen supervised segmentation by combining localisation-preserving encoder–decoder designs with attention or transformer-based global context modelling.

Complementary work explores data and label efficiency through semi-supervised learning, leveraging unlabelled endoscopic video to reduce annotation overhead [11], [12]. Representative strategies include self-supervision specific to medical imaging and consistency regularised semi-supervision, exemplified by Mean Teacher [13], cross-consistency training [14] and cross-pseudo supervision [15]. Collectively, these methods improve label efficiency by enforcing prediction consistency and exploiting unlabelled data. Notwithstanding advancements, two challenges remain: scalable modelling of long-range dependencies with favourable runtime and memory costs, and a principled integration of self-supervised representation learning with semi-supervised learning such that representation learning and label-efficient optimisation are mutually reinforcing. We target both by replacing convolutional blocks with visual state-space modules [16], [17] for efficient long-range modelling, and by unifying contrastive self-supervision with cross pseudo-label supervision under a single objective. Although recent work such as *Semi-Mamba-UNet* [18] has demonstrated the potential of Mamba-based semi-supervised medical segmentation, robotic and endoscopic surgical scenes introduce distinct challenges, including RGB surgical video appearance, instrument boundary ambiguity, specular reflection, occlusion, and computational constraints for surgical perception. These challenges motivate MAC-S^{3}Robo as a Mamba-driven contrastive semi-supervised framework tailored to robotic and endoscopic surgical segmentation.

In this paper, we present MAC-S^{3}Robo, a Mamba-based U-shaped segmentation framework for surgical robotics that integrates visual state-space (VSS) blocks, contrastive self-supervision, and dual-branch cross semi-supervision. Our contributions are fourfold:

1) We propose MAC-S^{3}Robo, a Mamba-driven contrastive semi-supervised segmentation framework tailored to robotic and endoscopic surgical segmentation, using dual Mamba-based U-shaped branches with visual state-space (VSS) modules.
2) We provide a computational cost analysis of the segmentation backbone, showing that MambaUNet requires fewer FLOPs and parameters than TransUNet and SwinUNet while maintaining comparable peak memory usage.
3) We formulate a unified objective that combines labelled supervision, cross pseudo-label supervision on unlabelled frames, and pixel-level contrastive self-supervision to improve label efficient surgical segmentation.
4) On the EndoVis 2017 robotic instrument segmentation benchmark [4] and EndoVis 2018 robotic scene segmentation benchmark [19], MAC-S^{3}Robo achieves the best mean Dice among the evaluated semi-supervised baselines at 10% labelled data and maintains consistent gains across labelled ratios from 10% to 90%.

II. Related Work

A. Segmentation Network Architecture

Image semantic segmentation has been a primary research focus in recent years, with an emphasis on developing deep learning networks for accurate and robust segmentation. In medical imaging, CNN-based models such as FCN [20], DeepLab [21], U-Net [5], and V-Net [22] have driven progress. Performance has been further enhanced by 3D CNNs [23], atrous CNNs [24], residual connections [25], attention mechanisms [9], and densely connected architectures [26], yielding state-of-the-art results on CT, MRI, and ultrasound tasks [5], [22]. In parallel, self-attention [27], notably the Vision Transformer (ViT) [28], has been adopted for dense prediction, exemplified by Swin Transformer [29] and SegFormer [30]. Within medical image analysis, many studies integrate conventional CNN techniques with ViT, yielding architectures such as TransUNet [10] and Swin-UNet [31].

B. Network Training Strategy

1) Semi-Supervised Learning

Semi-supervised learning (SSL) is now a standard approach for alleviating the shortage of labelled data in medical imaging. Foundational methods include temporal ensembling [32], Mean Teacher [13], and its uncertainty-aware variant [33]. More recent work employs strategies based on pseudo labels such as cross-pseudo supervision [15] to enforce consistency on unlabelled data. The central idea is to make predictions consistent under perturbations of the same sample and, in some cases, the network.

2) Contrastive Self-Supervised Learning

Contrastive learning is a form of self-supervised learning for acquiring discriminative representations by bringing positive pairs closer and separating negative pairs [34], [35]. In medical image analysis, contrastive objectives have been used to strengthen feature learning under limited annotations [36], [37]. In MAC-S^{3}Robo, this idea is used as a pixel-level auxiliary objective within the dual-branch semi-supervised framework, so that representation learning and pseudo-label based optimisation are encouraged jointly. Recent work such as *Semi-Mamba-UNet* [18] has explored Mamba-based

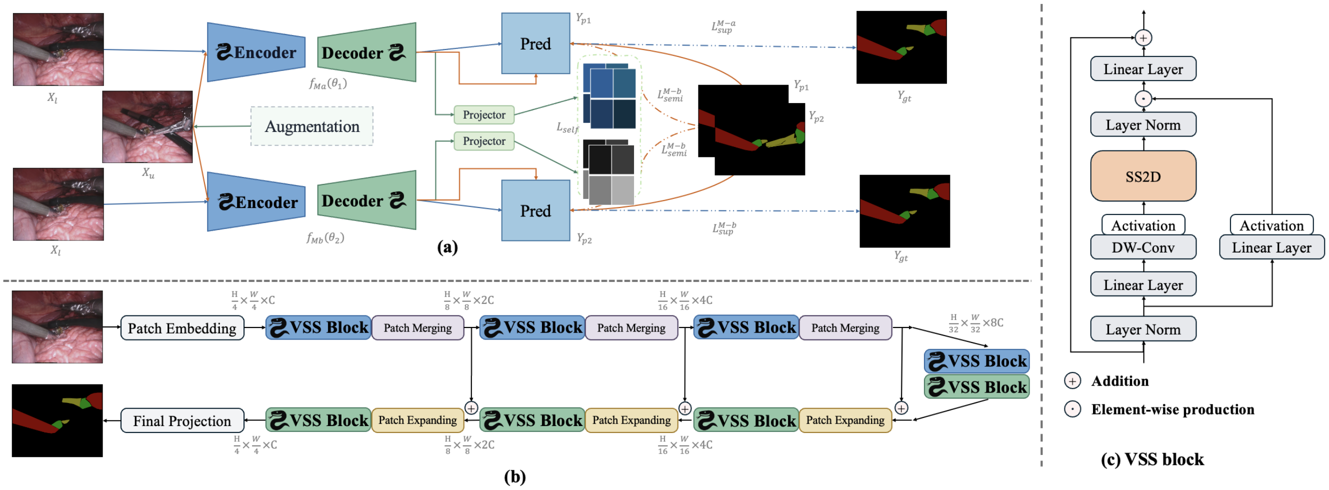

FIGURE 1. Overview of MAC-S^{3}Robo. (a) The proposed framework consists of a Mamba-based encoder–decoder network with semi-supervised and self-supervised learning for surgical robot segmentation. (b) VSS-based encoder–decoder network blocks. (c) VSS block.

semi-supervised medical image segmentation with pseudo-label exchange and pixel-level contrastive learning. In this paper, we focus on RGB robotic and endoscopic surgical segmentation and develop MAC-S^{3}Robo with dual Mamba-based branches in the EndoVis surgical benchmark setting.

C. Surgical Robot Segmentation

Robotic instrument segmentation has been studied primarily under the EndoVis 2017 Robotic Instrument Segmentation benchmark [4] and the EndoVis 2018 Robotic Scene Segmentation benchmark [19], which define binary instrument masks, part-level masks (shaft/wrist/jaws), instrument type categories, and more recently, multi-class scene parsing that includes both devices and anatomy. While these benchmarks have standardised tasks and evaluation protocols, a practical limitation is annotation scale and cost: pixel-level masks must be curated frame by frame and remain available only for relatively small subsets compared with clinical video archives. The EndoVis challenge reports [4], [19] and the broader surgical data science literature [1], [3], [38] emphasise the expense and heterogeneity of annotation. Consequently, recent studies have sought to reduce reliance on dense annotations by exploiting unlabelled endoscopic video, including self-supervised endoscopic representation learning [11], [12] and semi-supervised consistency methods such as Mean Teacher [13] and cross pseudo supervision [15].

Early EndoVis submissions were dominated by supervised CNN encoder–decoder models, including U-Net-style variants [5], [6]. Later work explored temporal consistency, stronger backbones, lightweight designs, and data augmentation to improve robustness in surgical video segmentation [4].

III. Approach

The architecture of MAC-S^{3}Robo is illustrated in Fig. 1. Motivated by visual state-space modelling and Mamba-based semi-supervised segmentation [16]–[18], MAC-S^{3}Robo adopts a dual Mamba-based U-Net architecture with cross pseudo-label supervision and a self-supervised contrastive objective for robotic and endoscopic surgical scenes. In this SSL task, we denote by $\mathbf{L}$ a small labelled set, by $\mathbf{U}$ a larger unlabelled set, and by $\mathbf{T}$ a held-out test set, with no overlap between them. A labelled mini-batch is written as $(\boldsymbol{X}_1, \boldsymbol{Y}_{\mathrm{gt}}) \in \mathbf{L}$, a test batch as $(\boldsymbol{X}_{\mathrm{t}}, \boldsymbol{Y}_{\mathrm{gt}}) \in \mathbf{T}$, and an unlabelled batch as $\boldsymbol{X}_{\mathrm{u}} \in \mathbf{U}$, where $\boldsymbol{X} \in \mathbb{R}^{h\times w\times c}$ denotes a 2D RGB image. Two segmentation networks $f_{\mathrm{Ma}}(\cdot;\theta_1)$ and $f_{\mathrm{Mb}}(\cdot;\theta_2)$ (Mamba-a / Mamba-b) produce per-pixel class probability maps $\boldsymbol{p}^{(a)}$ and $\boldsymbol{p}^{(b)}$. Pseudo-labels are obtained via pixel-wise argmax:

$$\boldsymbol{Y}_{\mathrm{p1}} = \arg\max_{c} \boldsymbol{p}^{(a)}(\boldsymbol{X})_c, \qquad \boldsymbol{Y}_{\mathrm{p2}} = \arg\max_{c} \boldsymbol{p}^{(b)}(\boldsymbol{X})_c, \tag{1}$$

where c indexes the class dimension. On unlabelled data, branch a is trained against $\boldsymbol{Y}_{\mathrm{p2}}$ and branch b against $\boldsymbol{Y}_{\mathrm{p1}}$ (cross-pseudo supervision), while on labelled data both branches are supervised by $\boldsymbol{Y}_{\mathrm{gt}}$. Final evaluation is performed by comparing the predicted masks $\boldsymbol{Y}_{\mathrm{pred}}$ to $\boldsymbol{Y}_{\mathrm{gt}}$ on $\mathbf{T}$.

A. State Space Model

Mamba [16], [39] builds on a state-space model (SSM) that describes sequence dynamics using a linear ordinary differential equation (ODE). In this continuous-time formulation, $h'(t)$ denotes the time derivative of the hidden state $h(t) \in \mathbb{R}^N$, and the model defines the mapping $x(t) \in \mathbb{R}^T \mapsto y(t) \in \mathbb{R}^T$ for an input of length T:

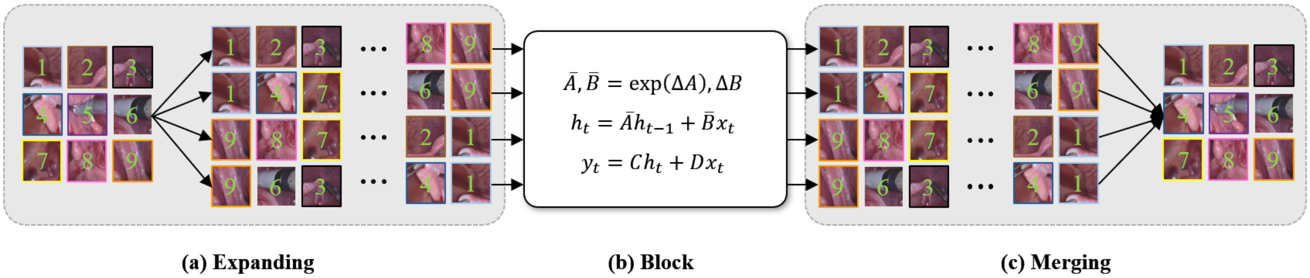

FIGURE 2. **Illustration of the expanding and merging scanning scheme and the visual state-space block.**

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad y(t) = \mathbf{C}h(t), \tag{2}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{B} \in \mathbb{R}^{N\times 1}$, and $\mathbf{C} \in \mathbb{R}^{1\times N}$ are the state, input, and output matrices, respectively. For discrete-time implementation, Mamba applies the zero-order hold (ZOH) discretisation, rewriting Eq. (2) as:

$$h_t = \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t, \quad y_t = \mathbf{C}h_t. \tag{3}$$

The discretisation parameters are computed as in Eq. (4), yielding a discrete-time SSM update that matches the continuous formulation over the sampling interval Δ.

$$\overline{\mathbf{A}} = \exp(\mathbf{A}\Delta), \quad \overline{\mathbf{B}} = (\mathbf{A}\Delta)^{-1}(\exp(\mathbf{A}\Delta) - \mathbf{I})\mathbf{B}\Delta. \tag{4}$$

In the discrete formulation, Mamba models the matrix $\overline{\mathbf{A}}$ as the state-transition parameter and $\overline{\mathbf{B}}$ and $\mathbf{C}$ as projection parameters. The output $\mathbf{y}$ is then computed from a length-T input sequence $\mathbf{x}$ using the structured convolutional kernel $\overline{\mathbf{K}}$, as detailed in Eq. (5).

$$\overline{\mathbf{K}} = (\mathbf{C}\overline{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}}\overline{\mathbf{B}}, \ldots, \mathbf{C}\overline{\mathbf{A}}^{T-1}\overline{\mathbf{B}}), \qquad \mathbf{y} = \mathbf{x} * \overline{\mathbf{K}}. \tag{5}$$

B. Mamba-Based U-Net Architecture

The architecture follows a U-shaped encoder–decoder with a 4×4 patch embedding front end, identity skip connections at matched scales, and a final projection head. Unlike the symmetric design popularised by Swin-UNet [31], we allocate depth and capacity asymmetrically across stages. Given an RGB image $x \in \mathbb{R}^{H\times W\times 3}$, the patch embedding partitions x into 4×4 non-overlapping patches and projects each patch to C channels (default $C = 96$), producing $x' \in \mathbb{R}^{\frac{H}{4}\times\frac{W}{4}\times C}$. Layer normalisation is applied to x' before entering the encoder.

The encoder comprises four stages with patch merging at the end of the first three, reducing spatial resolution as $\frac{H}{4}\times\frac{W}{4} \to \frac{H}{8}\times\frac{W}{8} \to \frac{H}{16}\times\frac{W}{16} \to \frac{H}{32}\times\frac{W}{32}$, while increasing channel width as $C \to 2C \to 4C \to 8C$. Each encoder stage stacks two VSS blocks, while the decoder uses two blocks per stage except the final stage which uses one. The decoder restores spatial detail by applying patch expansion at the beginning of the last three stages, bringing the feature maps back through $\frac{H}{16}\times\frac{W}{16} \to \frac{H}{8}\times\frac{W}{8} \to \frac{H}{4}\times\frac{W}{4}$, with channel widths $8C \to 4C \to 2C \to C$. After each expansion, the corresponding encoder features are added via an element-wise identity skip, and the result is further processed by VSS blocks. A final projection head upsamples by a factor of 4 and maps to class logits, yielding predictions at the original image resolution.

C. Objective

The training objective of MAC-S^{3}Robo is to minimize a total loss given by the sum of the supervised loss $\mathcal{L}_{\text{sup}}$, the self-supervised contrastive loss $\mathcal{L}_{\text{self}}$, and the semi-supervised loss $\mathcal{L}_{\text{semi}}$. We denote this sum by $\mathcal{L}_{\text{sum}}$, as shown in Eq. (6). Two optimizers then minimize the total loss by updating the parameters of the two segmentation networks ($f_{Ma}(\theta_1)$, $f_{Mb}(\theta_2)$) and the two projectors ($p^{M-a}(\cdot)$, $p^{M-b}(\cdot)$) in the dual Mamba-based UNet.

$$\mathcal{L}_{\text{sum}} = \overbrace{\underbrace{\mathcal{L}_{\text{sup}}^{\text{M-a}} + \mathcal{L}_{\text{sup}}^{\text{M-b}}}_{\text{sup}}}^{\text{labelled}} + \overbrace{\underbrace{\lambda_1\mathcal{L}_{\text{self}}}_{\text{self}} + \underbrace{\lambda_2\left(\mathcal{L}_{\text{semi}}^{\text{M-a}} + \mathcal{L}_{\text{semi}}^{\text{M-b}}\right)}_{\text{semi}}}^{\text{unlabelled}}. \tag{6}$$

where $\mathcal{L}_{\text{sup}}$ denotes the supervised loss on the labelled training set, $\mathcal{L}_{\text{semi}}$ the semi-supervised loss on the unlabelled training set, and $\mathcal{L}_{\text{self}}$ the self-supervised contrastive loss on unlabelled data. λ_1 and λ_2 (initial values set as 1 and 0.1) are scheduled with a ramp-up function [32] applied solely to the unlabelled training set, enabling MAC-S^{3}Robo to initialize from the labelled set and then gradually shift toward learning from the unlabelled set. The supervision loss $\mathcal{L}_{\text{sup}}$ is the sum of the per-network losses $\mathcal{L}_{\text{sup}}^{M-a}$ and $\mathcal{L}_{\text{sup}}^{M-b}$ on labelled pairs $(\boldsymbol{X}_l, \boldsymbol{Y}_l) \in \mathbf{L}$, illustrated as:

$$\mathcal{L}_{\text{sup}}^{\text{M-a}} = \text{CE}(Y_l, p_{\text{sup}}^{M-a}) + \text{Dice}(Y_l, p_{\text{sup}}^{M-a}),$$
$$\mathcal{L}_{\text{sup}}^{\text{M-b}} = \text{CE}(Y_l, p_{\text{sup}}^{M-b}) + \text{Dice}(Y_l, p_{\text{sup}}^{M-b}),$$

where CE and $Dice$ denote the cross-entropy and Dice coefficient losses, respectively. The self-supervised and semi-supervised losses $\mathcal{L}_{\text{self}}$ and $\mathcal{L}_{\text{semi}}$ are defined in Eq. (8) and Eq. (9), respectively.

FIGURE 3. Qualitative comparison of segmentation results on representative EndoVis examples. Masks are shown as transparent overlays on raw images, with column labels identifying each method.

D. Contrastive Self-Supervision

To stabilise representations on unlabelled frames within our dual-branch framework, we attach lightweight projection heads $p^{M-a}(\cdot), p^{M-b}(\cdot)$ to the last decoder stage of $f_{M_a}(\theta_1)$ and $f_{M_b}(\theta_2)$, respectively. Each projector maps per-pixel features to ℓ_2-normalised embeddings. The contrastive term is optimised only on unlabelled data and is combined with the supervised and semi-supervised terms in the overall objective L_{sum} (Eq. (6)).

Let $\mathbf{z}^{(a)}, \mathbf{z}^{(b)} \in \mathbb{R}^{H\times W\times d}$ denote the per-pixel embeddings from the two branches for an unlabelled input, with $\|\mathbf{z}_i^{(a)}\|_2 = \|\mathbf{z}_i^{(b)}\|_2 = 1$ at pixel index $i \in \Omega$. We treat the same spatial location across branches as a positive pair, and take other locations within the mini-batch as negatives. Using cosine similarity $\text{sim}(\mathbf{u},\mathbf{v}) = \mathbf{u}^\top\mathbf{v}$ with temperature $\tau > 0$, the one-way InfoNCE objective [34] from branch a to branch b is

$$\ell_i^{a\to b} = -\log \frac{\exp\big(\text{sim}(\mathbf{z}_i^{(a)}, \mathbf{z}_i^{(b)})/\tau\big)}{\sum_{j\in\mathcal{N}} \exp\big(\text{sim}(\mathbf{z}_i^{(a)}, \mathbf{z}_j^{(b)})/\tau\big)}, \tag{7}$$

where $\mathcal{N}$ collects negative indices from the current mini-batch (excluding i). Symmetrically, $\ell_i^{b\to a}$ is defined by swapping the roles of a and b. Averaging over all pixels yields the self-supervised contrastive term:

$$L_{\text{self}} = \frac{1}{|\Omega|}\sum_{i\in\Omega}\Big(\ell_i^{a\to b} + \ell_i^{b\to a}\Big). \tag{8}$$

This self-supervision encourages cross-branch agreement at the pixel level while repelling disparate locations, thereby improving the label efficiency of semi-supervised training. In the overall loss, $\mathcal{L}_{\text{sum}}$ is defined in Eq. (6), where λ_1 and λ_2 are scheduled on unlabelled data. The cross pseudo-supervision term $\mathcal{L}_{\text{semi}}$ follows Eq. (9).

E. Cross Semi-Supervision

Inspired by [15], [40], we introduce a semi-supervised framework guided by cross pseudo-label supervision for the Mamba-based UNet. Two Mamba-based segmentation branches share the same architecture but are initialized with different parameters. For the same input image $\boldsymbol{X}$, the pseudo-labels are $\boldsymbol{Y}_{\text{p1}} = \arg\max_c f_{\text{Ma}}(\boldsymbol{X};\theta_1)_c$ and $\boldsymbol{Y}_{\text{p2}} = \arg\max_c f_{\text{Mb}}(\boldsymbol{X};\theta_2)_c$, where c indexes the class dimension, and θ_1 and θ_2 denote the parameters of the two branches. The prediction from one branch is used as pseudo-label supervision for the other branch.

The semi-supervised loss for the two Mamba-based branches is calculated using cross-entropy CE as:

$$\begin{aligned} \mathcal{L}_{\text{semi}}^{\text{M-a}} &= \text{CE}\big(\boldsymbol{Y}_{\text{p2}}, f_{\text{Ma}}(\boldsymbol{X};\theta_1)\big), \\ \mathcal{L}_{\text{semi}}^{\text{M-b}} &= \text{CE}\big(\boldsymbol{Y}_{\text{p1}}, f_{\text{Mb}}(\boldsymbol{X};\theta_2)\big). \end{aligned} \tag{9}$$

IV. Experiments and Results

A. Dataset

We use the EndoVis 2017 robotic instrument segmentation release [4] and the EndoVis 2018 robotic scene segmentation release [19]. The 2017 release contains ten stereo sequences acquired with a da Vinci Xi system at 1280×1024 (SXGA), and the 2018 release contains nineteen stereo sequences captured with da Vinci X/Xi systems at the same resolution.

B. Implementation Details

Our code was developed on Ubuntu 20.04 with Python 3.8.8, PyTorch 1.10, and CUDA 11.3, running on a single NVIDIA GeForce RTX 3090 GPU and an Intel Core i9-10900K CPU. The end-to-end runtime per experiment averaged approximately 3.5 hours, including data transfer, training, and inference. The dataset was processed for 2D

TABLE 1. Semi-supervised surgical robot segmentation on EndoVis 2017. All models are trained with 10% labelled data, with the remaining frames used as unlabelled data. The best validation checkpoint is evaluated on the test set.

Method	Dice↑	ACC↑	PRE↑	SEN↑	SPE↑
MT [13]	0.471	0.881	0.504	0.441	0.941
UAMT [33]	0.504	0.889	0.541	0.472	0.945
CPS [15]	0.534	0.895	0.571	0.502	0.949
ICT [41]	0.476	0.878	0.491	0.461	0.935
FixMatch [42]	0.526	0.885	0.521	0.531	0.933
TriSegNet [43]	0.558	0.901	0.601	0.521	0.953
MixSegNet [44]	0.574	0.904	0.611	0.541	0.953
SegMatch [45]	0.566	0.898	0.581	0.552	0.946
Surgivisor [46]	0.576	0.901	0.591	0.561	0.947
MAC-S^{3}Robo	**0.598**	**0.909**	**0.638**	**0.562**	**0.957**

TABLE 2. Semi-supervised surgical robot segmentation on EndoVis 2018. All models are trained with 10% labelled data, with the remaining frames used as unlabelled data. The best validation checkpoint is evaluated on the test set.

Method	Dice↑	ACC↑	PRE↑	SEN↑	SPE↑
MT [13]	0.337	0.866	0.413	0.285	0.945
UAMT [33]	0.369	0.866	0.424	0.327	0.939
CPS [15]	0.462	0.885	**0.526**	0.412	**0.949**
ICT [41]	0.412	0.858	0.409	0.416	0.918
FixMatch [42]	0.438	0.878	0.491	0.395	0.944
TriSegNet [43]	0.488	0.874	0.478	0.498	0.926
MixSegNet [44]	0.509	0.882	0.509	0.509	0.933
SegMatch [45]	0.528	0.878	0.494	0.566	0.921
Surgivisor [46]	0.541	0.884	0.516	0.569	0.927
MAC-S^{3}Robo	**0.549**	**0.887**	0.525	**0.575**	0.929

image segmentation. MAC-S^{3}Robo was trained for 30,000 iterations with a batch size of 24 using SGD with a learning rate of 0.01, momentum of 0.9, and weight decay of 0.0001. The network was evaluated on the validation set every 200 iterations, and weights were saved only when validation performance improved over the previous best. The same settings were applied to all baseline methods without modification.

We follow the official inter-sequence protocol for both benchmarks. For EndoVis 2017, we train on eight sequences (8×225 = 1,800 frames) and evaluate on the remaining two full-length sequences (2×300 = 600 frames). For EndoVis 2018, we train on fifteen sequences and evaluate on four sequences, which in our setup correspond to 2,235 training frames and 997 test frames. To assess label efficiency, we also simulate limited supervision by randomly subsampling the available annotations at 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% of the training masks, while keeping the remaining frames unlabelled for semi-supervised learning. The reported quantitative values correspond to this stated labelled data ratio protocol and are not intended to represent confidence intervals over repeated independent training runs.

C. Metrics

The direct comparisons between MAC-S^{3}Robo and the baseline methods use a comprehensive set of evaluation metrics: Dice, accuracy, precision, sensitivity, and specificity (higher is better). Because the dataset is multi-class, we report the mean value of each metric across classes. The full set of evaluation metrics is reported for all comparisons with the baselines.

D. Segmentation Backbone Network

For the experiments, we employ ViT and CNN layers within a unified encoder–decoder segmentation architecture to facilitate a fair comparison. At each encoder or decoder level, the CNN block consists of two consecutive 3×3 convolutions, dropout, and batch normalisation. Alternatively, the ViT block uses layer normalisation, shifted-window multi-head self-attention, residual connections, and a multi-layer perceptron. This design yields pure CNN- and pure ViT-based UNet structures [5], [31]. For our proposed framework, we replace UNet's convolutional blocks with Mamba-based visual state-space (VSS) blocks, resulting in a Mamba-based U-Net backbone. Each stage contains VSS blocks derived from the selective state-space model (see Sec. III-A, Eq. (2)–(5)), while preserving the multi-scale encoder–decoder design, identity skip connections, and the final projection head of the UNet. This design leverages the discrete-time SSM with ZOH discretisation and the associated structured convolutional kernel to model long-range dependencies and global context within the same U-shaped architecture.

Table 5 compares the computational cost of representative segmentation backbones. MambaUNet requires 3.5211G FLOPs and 15.4780M parameters, with 1.5GB peak memory, indicating lower computational cost than TransUNet and SwinUNet in FLOPs and parameter count while maintaining comparable peak memory usage. In practice, the expected throughput is likely to be in the range of several hundred FPS for 512×512 inputs, and can be higher with FP16 inference and batch processing. This comparison supports the use of the Mamba-based backbone under practical computational constraints.

E. Baseline Methods

We compare MAC-S^{3}Robo against strong semi-supervised segmentation baselines including MT (Mean Teacher) [13]; UAMT [33]; CPS [15]; ICT [41]; FixMatch [42]; TriSegNet [43]; MixSegNet [44]; SegMatch [45]; Surgivisor [46]. All listed baselines are included in the main 10% labelled data comparison. For the labelled data ratio sensitivity analysis, we use selected representative baselines to keep the trend analysis compact and readable.

TABLE 3. Compact labelled data ratio sensitivity analysis on EndoVis 2017 and EndoVis 2018. Mean Dice is reported using selected representative semi-supervised baselines.

Dataset	Method	10%	20%	30%	40%	50%	60%	70%	80%	90%
EndoVis@MICCAI'17	MT	0.471	0.531	0.581	0.626	0.666	0.701	0.726	0.746	0.756
	UAMT	0.504	0.559	0.609	0.649	0.687	0.721	0.745	0.763	0.773
	CPS	0.534	0.586	0.629	0.667	0.701	0.731	0.756	0.774	0.786
	FixMatch	0.526	0.576	0.619	0.656	0.691	0.721	0.747	0.767	0.778
	MAC-S^{3}Robo	**0.598**	**0.651**	**0.691**	**0.725**	**0.754**	**0.777**	**0.794**	**0.803**	**0.807**
EndoVis@MICCAI'18	MT	0.337	0.417	0.487	0.547	0.602	0.652	0.697	0.727	0.742
	UAMT	0.369	0.449	0.514	0.569	0.619	0.669	0.709	0.744	0.758
	CPS	0.462	0.532	0.591	0.641	0.685	0.725	0.761	0.781	0.791
	FixMatch	0.438	0.508	0.563	0.613	0.658	0.698	0.733	0.758	0.772
	MAC-S^{3}Robo	**0.549**	**0.614**	**0.661**	**0.701**	**0.735**	**0.763**	**0.784**	**0.796**	**0.801**

TABLE 4. Ablation on EndoVis@MICCAI'17 with 10% labelled data.

Backbone / Variant	Contrastive Self-Sup	Semi-Sup	Dice↑	ACC↑	PRE↑	SEN↑	SPE↑
CNN baseline			0.397	0.869	0.442	0.361	0.938
CNN + Cross-Sup.		✓	0.476	0.881	0.503	0.452	0.939
CNN + Contrastive + Cross-Sup.	✓	✓	0.517	0.887	0.531	0.503	0.939
ViT baseline			0.432	0.876	0.481	0.392	0.942
ViT + Cross-Sup.		✓	0.508	0.891	0.552	0.471	0.948
ViT + Contrastive + Cross-Sup.	✓	✓	0.542	0.894	0.563	0.522	0.945
Mamba-UNet			0.466	0.884	0.521	0.421	0.947
Mamba-UNet + Cross-Sup.		✓	0.535	0.895	0.572	0.503	0.949
MAC-S^{3}Robo	✓	✓	**0.598**	**0.909**	**0.638**	**0.562**	**0.957**

TABLE 5. Computational cost comparison of segmentation backbones.

Backbone	TransUNet	SwinUNet	MambaUNet
FLOPs (G)	19.6787	8.6929	3.5211
Params (M)	90.4423	41.3415	15.4780
Peak memory (GB)	2.8	1.4	1.5

F. Qualitative Results

Figure 3 presents six randomly selected raw images from two datasets together with the corresponding predictions and the provided ground truth for all baseline methods and MAC-S^{3}Robo. Pixel-level regions on the test set are color-coded as multi-class segmentation.

G. Quantitative Results

Table 1 and Table 2 summarize a direct comparison between MAC-S^{3}Robo and other SSL methods at a labelled/total data ratio of 10%, reporting the following evaluation metrics (higher is better): Dice, Accuracy, Precision, Sensitivity, Specificity. Table 3 reports a compact labelled data ratio sensitivity analysis using selected representative semi-supervised baselines, and Fig. 4 further visualises the corresponding Mean Dice trends on EndoVis 2017 and EndoVis 2018. The best entries in Table 1, Table 2, and Table 3 are typeset in **Bold**. All quantitative results are reported under the stated split and labelled data ratio protocol, without repeated independent training runs.

FIGURE 4. Mean Dice versus labelled data ratio on EndoVis 2017 and EndoVis 2018. The curves visualise the results reported in Table 3.

H. Ablation Study

Table 4 reports a controlled ablation along three axes: the Mamba backbone, contrastive self-supervision, and semi-supervised cross-supervision. All experiments use only 10% of the labelled data.

a: Mamba (backbone effect).

With cross-supervision and contrastive pre-training enabled, MAC-S^{3}Robo attains the best overall performance (Dice 0.598). The gain over the ViT counterpart is $+0.056$ Dice and the gain over the CNN counterpart is $+0.081$. Accuracy and precision show similar margins relative to ViT and CNN (about $+0.015$ and $+0.022$ in accuracy, $+0.075$ and $+0.107$ in precision). Even without self or semi supervision, a Mamba-UNet baseline exceeds both CNN and ViT baselines (Dice $+0.069$ and $+0.034$), indicating a backbone advantage from selective state-space modelling.

b: Semi-supervised cross-supervision (consistency effect).

Turning on cross-supervision improves Dice for every backbone: $+0.079$ for CNN ($0.397 \rightarrow 0.476$), $+0.076$ for ViT ($0.432 \rightarrow 0.508$) and $+0.069$ for Mamba ($0.466 \rightarrow 0.535$). The dominant gains are on sensitivity (roughly $+0.079$ to $+0.091$), while specificity remains essentially unchanged (variation within ± 0.006). This indicates better recall without compromising negative class reliability.

c: Contrastive self-supervision ($\mathcal{L}_{con}$).

On top of cross-supervision, contrastive initialisation yields further Dice gains of $+0.041$ for CNN ($0.476 \rightarrow 0.517$), $+0.034$ for ViT ($0.508 \rightarrow 0.542$) and $+0.063$ for Mamba ($0.535 \rightarrow 0.598$). Sensitivity increases by about $+0.051$ for CNN and ViT and by $+0.059$ for Mamba, with specificity stable (a small -0.003 on ViT and a positive shift on Mamba).

V. Conclusion

In this paper, we present MAC-S^{3}Robo, a Mamba-based U-shaped architecture that combines contrastive self-supervision for representation learning with cross-supervision for semi-supervised surgical robotic instrument and scene segmentation. By leveraging Mamba-based sequence modelling and contrastive representation learning, MAC-S^{3}Robo improves segmentation performance over the evaluated baselines under limited labelled data. The added inference analysis further provides an initial assessment of its computational feasibility for surgical robotic perception. Future work will explore extending MAC-S^{3}Robo to multimodal surgical datasets and improving computational efficiency for clinically oriented robotic platforms.

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